

# VPA IB DP CS — IA Guide and Support

## IA Guide and Support 内部评估指南 Nèibù Pínggū Zhǐnán

Weight: 30% SL / 20% HL | Hours: ~34 class hours + independent study

### 1. What is the IA?

The Internal Assessment is a software development project that solves a real problem for a real client. It is not a hypothetical exercise — the client must exist, the problem must be genuine, and the solution must be implemented and evaluated.

### 2. IA Timeline (2-Year View)

| Date            | Milestone                 | Criterion | Deliverable                                     |
|-----------------|---------------------------|-----------|-------------------------------------------------|
| Year 1, Week 8  | Client identified         | A         | Client name, role, contact details              |
| Year 1, Week 16 | Problem confirmed         | A         | Written problem statement with client signature |
| Year 1, Week 18 | Success criteria drafted  | A         | 3–5 measurable criteria                         |
| Year 2, Week 1  | Planning complete         | A         | Full Criterion A with record of tasks started   |
| Year 2, Week 3  | Design complete           | B         | System diagrams, algorithms, test plan          |
| Year 2, Week 9  | Development core complete | C         | Working prototype with key features             |
| Year 2, Week 12 | Testing complete          | D         | Test table with evidence                        |
| Year 2, Week 14 | Evaluation complete       | E         | Client feedback, success criteria review        |
| Year 2, Week 15 | First full draft          | A–E       | Complete documentation for feedback             |
| Year 2, Week 17 | FINAL SUBMISSION          | A–E       | Upload to IBIS + video evidence                 |

### 3. Criterion A: Planning (6 marks)

#### A.1 — The Scenario (Context)

Describe the client's organisation and their role. Explain why the client needs a software solution.

Good example:

"Ms. Zhang is the librarian at Victoria Park Academy Shenzhen. She currently manages book loans using a paper logbook, which makes it difficult to track overdue books, identify popular titles, or generate reports for the administration. She has agreed to be my client for this project."

## A.2 — The Problem

State the problem clearly, in the client's own words where possible. The problem must be specific and solvable with software.

## A.3 — Rationale

Explain why a computational solution is appropriate. Consider alternatives and justify your choice.

## A.4 — Success Criteria

3–5 criteria that are specific, measurable, and testable.

Good example:

1. The system shall allow a book to be checked out in fewer than 30 seconds.
2. The system shall generate an overdue report automatically every Monday morning.
3. The system shall store at least 5,000 book records.
4. The system shall be usable by the librarian without technical training (intuitive interface).

## A.5 — Record of Tasks

A dated log of all activities undertaken. Must show evidence of consultation with the client.

| Date       | Task                                                       | Time   | Client Consultation?          |
|------------|------------------------------------------------------------|--------|-------------------------------|
| 2025-09-10 | Initial meeting with Ms. Zhang to discuss library problems | 30 min | Yes — meeting notes attached  |
| 2025-09-15 | Drafted success criteria; sent to client for approval      | 45 min | Yes — email response received |

## 4. Criterion B: Design (6 marks)

### B.1 — Record of Tasks (continued)

The record of tasks spans Criteria A and B. Show design-phase activities.

### B.2 — System Overview

A diagram showing the overall structure of your solution. Include:

Main modules/components

Data storage

User interface overview

Data flow between components

### B.3 — Algorithms

Design your key algorithms in pseudocode before coding. Show:

Input and output

Processing steps

Loops and conditions

Complex techniques (e.g., file handling, searching, sorting)

## B.4 — Test Plan

| Test # | Success Criterion   | Test Data                      | Expected Result                       | Type      |
|--------|---------------------|--------------------------------|---------------------------------------|-----------|
| 1      | Check out in <30s   | Valid book ID, valid member ID | Loan recorded; confirmation displayed | Normal    |
| 2      | Overdue report      | 5 overdue books, 3 on-time     | Report lists exactly the 5 overdue    | Normal    |
| 3      | Store 5,000 records | 5,000 book records loaded      | System remains responsive             | Boundary  |
| 4      | Usability           | Invalid book ID entered        | Error message; no crash               | Erroneous |

## 5. Criterion C: Development (12 marks)

### C.1 — Complex Techniques

To achieve high marks, you must demonstrate at least two complex techniques. Examples include:

File handling (reading/writing structured data)

Searching or sorting algorithms

Use of collections/arrays of records

Object-oriented programming (classes, inheritance)

Database connectivity (SQL)

API integration

Recursive algorithms

Multi-threading/concurrency

### C.2 — Code with Commentary

Include key sections of code with inline commentary explaining:

What the code does

Why you chose this approach

How it relates to your design

### C.3 — Video Evidence

A 5–7 minute video demonstrating your solution. The video must:

Show the program running

Demonstrate each success criterion being met

Include narration explaining what is happening

Be clear and well-paced

## 6. Criterion D: Evaluation (4 marks)

### D.1 — Testing Against Success Criteria

Show evidence that each success criterion was tested. Use screenshots, test data, and results.

### D.2 — Client Feedback

Include direct feedback from your client. This can be:

Written evaluation

Email response

Meeting notes

Audio/video recording

### D.3 — Future Improvements

Identify at least two realistic improvements that could be made with more time/resources.

## 7. Common IA Pitfalls

| Pitfall                         | Why It Fails                                    | How to Avoid                                                     |
|---------------------------------|-------------------------------------------------|------------------------------------------------------------------|
| Fake client                     | IB moderators can detect invented clients       | Choose a real person; document all contact                       |
| Trivial problem                 | Insufficient complexity for 12-mark Criterion C | Ensure problem requires multiple features and complex techniques |
| No client consultation evidence | Criterion A and D require proof of interaction  | Save all emails, meeting notes, photos                           |
| Success criteria too vague      | Cannot be tested or evaluated                   | Use numbers, times, quantities                                   |
| Video too long or unclear       | Moderator cannot verify functionality           | Script the video; keep to 7 minutes; show each criterion clearly |
| Design after coding             | Design must precede development                 | Complete design before writing code                              |

## 8. IA Exemplar Outline (Grade 7)

Client: PE Department Head

Problem: Managing sports equipment loans, tracking condition, generating maintenance alerts

Solution: Equipment inventory system with barcode scanning, loan tracking, automated maintenance scheduling

Complex Techniques:

File I/O with CSV parsing (structured data handling)

Binary search for equipment lookup (efficient searching)

Object-oriented design (Equipment, Loan, User classes)

## 9. EAL Support for IA

| Challenge                   | Support Provided                                                                  |
|-----------------------------|-----------------------------------------------------------------------------------|
| Writing in academic English | Bilingual templates; sentence starters; peer review with fluent English speakers  |
| Understanding command terms | Bilingual glossary of IA-specific terms; exemplar annotations                     |
| Client communication        | Permission to conduct consultations in L1; translation of key documents permitted |
| Technical vocabulary        | Bilingual code commentary template; pre-translated common phrases                 |

IA Final Submission Checklist:

- ☐ Criterion A: Planning (client, problem, rationale, success criteria, record of tasks)
- ☐ Criterion B: Design (record of tasks, system overview, algorithms, test plan)
- ☐ Criterion C: Development (complex techniques, commented code, video evidence)
- ☐ Criterion D: Evaluation (testing evidence, client feedback, future improvements)
- ☐ Word count checked (no maximum, but quality over quantity)
- ☐ Video uploaded and link working
- ☐ File naming follows IB requirements
- ☐ Coversheet completed
- ☐ Teacher has reviewed draft and given feedback
- ☐ Victoria Park Academy Shenzhen | IB DP Computer Science | IA Guide